

FREEDOM INTERNATIONAL SCHOOL
MATHEMATICS
APPLICATIONS OF DIFFERENTIATION
INCREASING AND DECREASING FUNCTIONS

CLASS 12

NAME:

1. The interval on which the function $f(x) = 2x^3 + 9x^2 + 12x - 1$ is decreasing in
a) $[-1, \infty)$ b) $[-2, -1]$ c) $(-\infty, 2]$ d) $[-1, 1]$
2. Which of the following functions is decreasing in $(0, \frac{\pi}{2})$?
a) $\sin 2x$ b) $\tan x$ c) $\cos x$ d) $\cos 3x$
3. The function $f(x) = \tan x - x$
a) always increases
b) always decreases
c) never increases
d) sometimes increases and sometimes decreases
4. The values of a for which the function $f(x) = \sin x - ax + b$ increases on $\mathbb{R}$ are
a) $(-\infty, 1]$ b) $(-\infty, -1)$ c) $(-\infty, 1]$ d) $(-\infty, 1)$
5. The function $f(x) = 4 \sin^3 x - 6 \sin^2 x + 12 \sin x + 100$ is strictly
a) decreases in $(\frac{-\pi}{2}, \frac{\pi}{2})$
b) decreasing in $(0, \frac{\pi}{2})$
c) decreasing in $(\frac{\pi}{2}, \pi)$
d) increases in $(\pi, \frac{3\pi}{2})$
6. On which of the following intervals is the function f given by
 $f(x) = x^{100} + \sin x - 1$ strictly decreasing?
a) $(0, 1)$ b) $(\frac{\pi}{2}, \pi)$ c) $(0, \frac{\pi}{2})$ d) none of these
7. The interval in which $y = x^2 e^{-x}$ is increasing is
a) $(-\infty, \infty)$ b) $(-2, 0)$ c) $(2, \infty)$ d) $(0, 2)$

8. The function $f(x) = xe^{1-x}$

- a) decreases in $(0,2)$
- b) increases in $(0,\infty)$
- c) strictly decreases in $(1,\infty)$
- d) strictly increases in $\left(\frac{1}{2}, 2\right)$

9. The function $f(x) = 2x^3 - 9x^2 + 12x + 99$ is monotonically decreases when

- a) $1 < x < 2$
- b) $x > 2$
- c) $x < 1$
- d) $x > 1$

10. If $f(x) = x^2 + ax + 5$ is increasing function in $[2,3]$ then the minimum value of a is

- a) -4
- b) -2
- c) 2
- d) 4

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MAXIMA AND MINIMA

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NAME:

1. Let the function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2x + \cos x$, then f

- a) has a minimum at $x = \pi$
- b) has a minimum at $x = 0$
- c) is a decreasing function
- d) is an increasing function

2. The function $f(x) = 2x^3 - 3x^2 - 12x - 4$, has

- a) two points of local maximum
- b) two points of local minimum
- c) one maxima and one minima
- d) no maxima or minima

3. At $x = \frac{5\pi}{6}$, $f(x) = 2\sin 3x + 3\cos 3x$ is

- a) maximum
- b) minimum
- c) zero
- d) neither maximum nor minimum

4. If x is real, then the minimum value of $x^2 - 8x + 17$ is

- a) -1
- b) 0
- c) 2
- d) 1

5. The smallest value of polynomial $x^3 - 18x^2 + 96x$ in $[0, 9]$ is

- a) 126
- b) 0
- c) 135
- d) 160

6. The maximum value of $\sin x \cdot \cos x$ is

- a) $1/4$
- b) $1/2$
- c) $\sqrt{2}$
- d) $2\sqrt{2}$

7. $\sin x + \cos x$ is maximum when $x =$

- a) $-\frac{\pi}{3}$
- b) 0
- c) $\frac{\pi}{3}$
- d) $\frac{\pi}{4}$

8. The maximum slope of the curve $y = -x^3 + 3x^2 + 9x - 27$ is
a) -1 b) 0 c) 2 d) 12

9. The maximum value of $\left(\frac{1}{x}\right)^x$ is

- a) e
- b) $(e)^e$
- c) $(e)^{1/e}$
- d) $\left(\frac{1}{e}\right)^{1/e}$

10. The function $f(x) = x^x$ has a stationary point at

- a) $x = e$
- b) $x = \frac{1}{e}$
- c) $x = 1$
- d) $x = \sqrt{e}$

11. The minimum value of the curve $y = xe^x$ is

- a) $\frac{1}{e}$ b) $\frac{-1}{e}$ c) e d) $-e$

12. The function $x^2 - 4x$, $x \in [0, 4]$ attains minimum value at $x =$

- a) 1 b) 0 c) 2 d) 4

13. The maximum slope of the curve $y = -x^3 + 3x^2 + 3x - 1$ is

- a) -1 b) 6 c) 2 d) 12

14. What is the maximum value of $\sin x + \cos x - \sqrt{2}$?

- a) $-\sqrt{2}$ b) $\sqrt{2}$ c) 0 d) 1

15. The maximum value of $f(x) = \sin(\sin x) \forall x \in \mathbb{R}$ is

- a) $\sin(1)$
- b) $-\sin(1)$
- c) $[-\sin(1), \sin(1)]$
- d) 1

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RATE MEASURE

CLASS 12

NAME:

1. The sides of an equilateral triangle are increasing at the rate of 2 cm/s. The rate at which the area increases, when side is 10 cm is
 - a) $10 \text{ cm}^2/\text{s}$
 - b) $\sqrt{3} \text{ cm}^2/\text{s}$
 - c) $10\sqrt{3} \text{ cm}^2/\text{s}$
 - d) $10/3 \text{ cm}^2/\text{s}$
2. A ladder 5 m long, standing on a horizontal floor, leans against a vertical wall. If the top of the ladder slides downwards at the rate of 10 cm/s then the rate at which the angle between the floor and the ladder is decreasing when lower end of the ladder is 2 m from the wall is
 - a) $1/10 \text{ rad/s}$
 - b) $1/20 \text{ rad/s}$
 - c) 20 rad/s
 - d) 10 rad/s
3. A cylindrical tank of radius 10 m is being filled with wheat at the rate of 314 cubic metres per hour. Then the depth of the wheat is increasing at the rate of
 - a) 1 m/h
 - b) 0.1 m/h
 - c) 1.1 m/h
 - d) 0.5 m/h
4. The rate of change of the area of a circle w.r.t its radius r at $r = 6 \text{ cm}$ is
 - a) $10\pi \text{ cm}$
 - b) $12\pi \text{ cm}$
 - c) $11\pi \text{ cm}$
 - d) $8\pi \text{ cm}$
5. The total revenue in rupees received from the sale of x units of a product is given by $R(x) = 3x^2 + 36x + 5$. The marginal revenue, when $x = 15$ is
 - a) 126
 - b) 90
 - c) 136
 - d) 96

6. If the radius of the circle is increasing at the rate of 0.5 cm/s, then the rate of increase of its circumference is
a) $\frac{2\pi}{3}$ cm/s b) $\frac{4\pi}{3}$ cm/s c) π cm/s d) $\frac{\pi}{3}$ cm/s
7. For the curve $y = -2x^3 + 5x$, if x increases at the rate of 2 units/sec, then at $x = 3$, the slope of the curve is changing at
a) decreasing at 72 units/sec
b) decreasing at -72 units/sec
c) changing at 7 units/sec
d) none of these
8. A swimming pool is to be drained for cleaning. If L represents the number of litres of water in the pool t seconds after the pool has been plugged off to drain and $L = 200(10 - t^2)$. How fast is the water draining out at the end of 5 seconds and what is the average rate at which the water flows out during the first 5 seconds?
a) 2000 L/s, 3000 L/s
b) 2000 L/s, 4000 L/s
c) 2000 L/s, 2000 L/s
d) 2000 L/s, 5000 L/s
9. The volume of a cube is increasing at the rate of $8 \text{ cm}^3/\text{s}$. How fast is the surface area increasing when the length of an edge is 12 cm?
a) $8/3 \text{ cm}^2/\text{s}$
b) $10/3 \text{ cm}^2/\text{s}$
c) $8 \text{ cm}^2/\text{s}$
d) $10 \text{ cm}^2/\text{s}$
10. The total cost $C(x)$ in rupees associated with the production of x units of an item is given by $C(x) = 13x^2 + 26x + 15$. Find the marginal cost, when 7 units are produced.
a) 206
b) 208
c) 210
d) 212